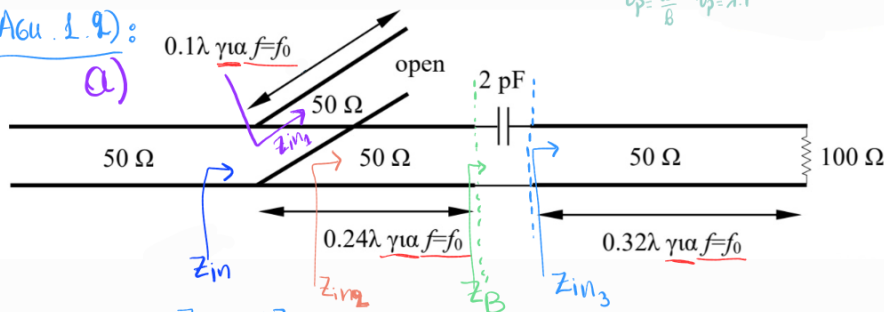


Αδ. 1.9):

Q)



$$v_p = \frac{\omega}{\beta} \quad v_p = \lambda \cdot f$$

$$\beta l = \frac{\omega}{v_p} \cdot l = \frac{2\pi f}{v_p} \cdot l$$

$$0.32\lambda = 0.32 \cdot \frac{v_p}{f_0} \quad \text{Αρα } \beta l = \frac{2\pi f}{v_p} \cdot 0.32 \frac{v_p}{f_0}$$

$$\Rightarrow \beta l = 0.32 \cdot 2\pi f / f_0$$

$$0.24\lambda \Rightarrow \beta l = 0.24 \cdot 2\pi f / f_0$$

$$0.1\lambda \Rightarrow \beta l = 0.1 \cdot 2\pi f / f_0$$

$$Z_{in3} = Z_0 \frac{Z_L + jZ_0 \tan(\beta l_3)}{Z_0 + jZ_L \tan(\beta l_3)} = 50 \frac{100 + j50 \tan(0.32 \cdot 2\pi f / f_0)}{50 + j100 \tan(0.32 \cdot 2\pi f / f_0)} = 50 \frac{2 + j \tan(0.32 \cdot 2\pi f / f_0)}{1 + j2 \tan(0.32 \cdot 2\pi f / f_0)}$$

$$Z_B = Z_{in3} + jX = Z_{in3} + j\left(-\frac{1}{2\pi f C}\right) = 50 \frac{2 + j \tan(0.32 \cdot 2\pi f / f_0)}{1 + j2 \tan(0.32 \cdot 2\pi f / f_0)} + j\left(-\frac{1}{4\pi f 10^{-12}}\right)$$

$$Z_{in2} = Z_0 \frac{Z_B + jZ_0 \tan(\beta l_2)}{Z_0 + jZ_B \tan(\beta l_2)} = 50 \frac{50 \frac{2 + j \tan(0.32 \cdot 2\pi f / f_0)}{1 + j2 \tan(0.32 \cdot 2\pi f / f_0)} + j\left(-\frac{1}{4\pi f 10^{-12}}\right) + j50 \tan(0.24 \cdot 2\pi f / f_0)}{50 + j\left(50 \frac{2 + j \tan(0.32 \cdot 2\pi f / f_0)}{1 + j2 \tan(0.32 \cdot 2\pi f / f_0)} + j\left(-\frac{1}{4\pi f 10^{-12}}\right)\right) \tan(0.24 \cdot 2\pi f / f_0)}$$

$$Z_{in1} \rightarrow Y_{in1}(s) = Y_0 \frac{Y_L + jY_0 \tan(\beta l_1)}{Y_0 + jY_L \tan(\beta l_1)} = \frac{1}{50} \cdot \frac{0 + j\frac{1}{50} \tan(0.1 \cdot 2\pi f / f_0)}{\frac{1}{50} + j \cdot 0} = \frac{j \tan(0.1 \cdot 2\pi f / f_0)}{50}$$

όπου $Y_0 = \frac{1}{Z_0}$

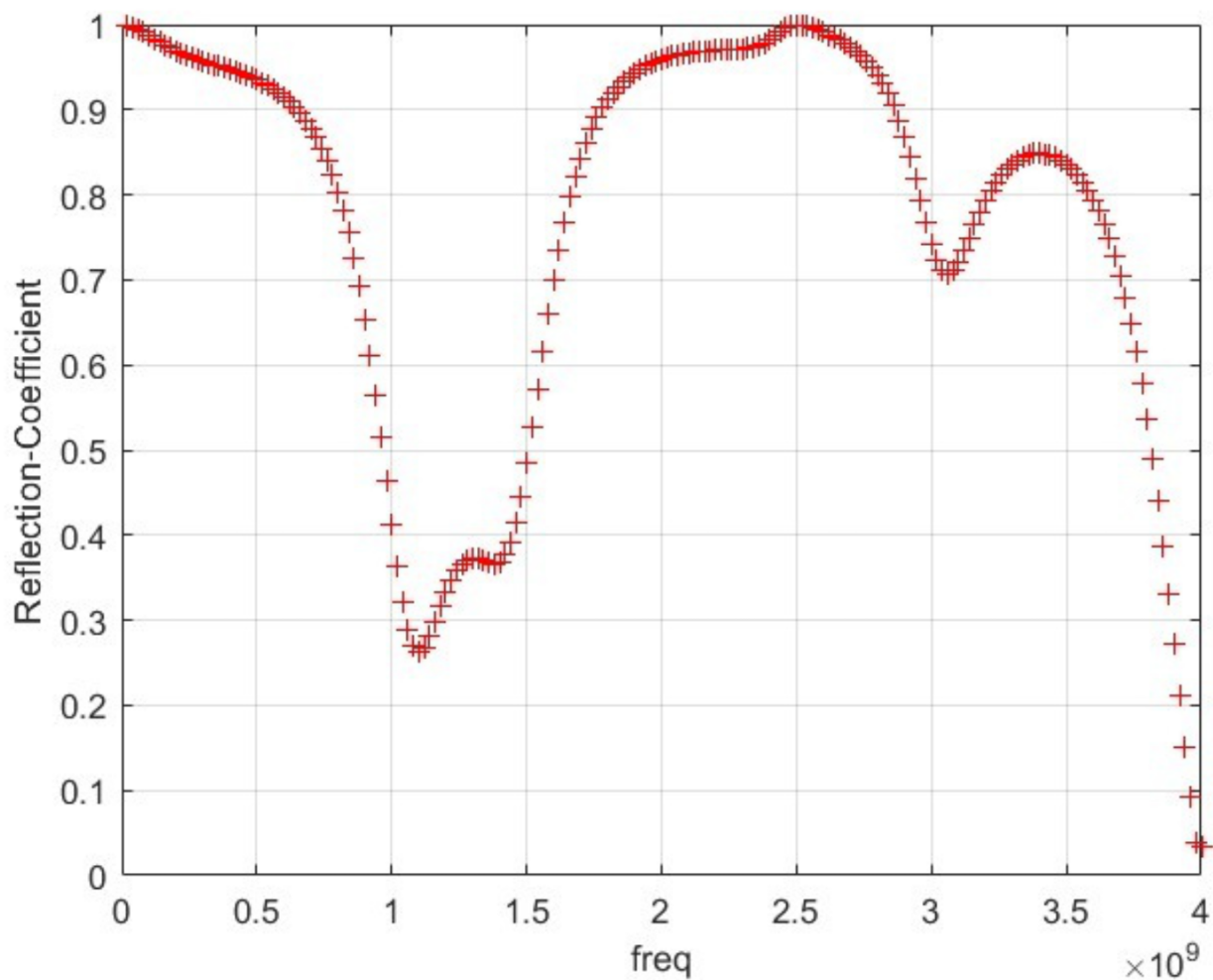
ή βάσει $Z_{in2} = Z_0 \frac{Z_L + jZ_0 \tan(\beta l_2)}{Z_0 + jZ_L \tan(\beta l_2)} = \frac{Z_L}{jZ_0 \tan(\beta l_2)} = \frac{Z_0}{j \tan(0.1 \cdot 2\pi f / f_0)}$

$$Z_{in} = Z_{in1} \parallel Z_{in2} = \frac{Z_{in1} \cdot Z_{in2}}{Z_{in1} + Z_{in2}} = \dots$$

Αρα $\left| \Gamma_{in} \right| = \left| \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \right|$

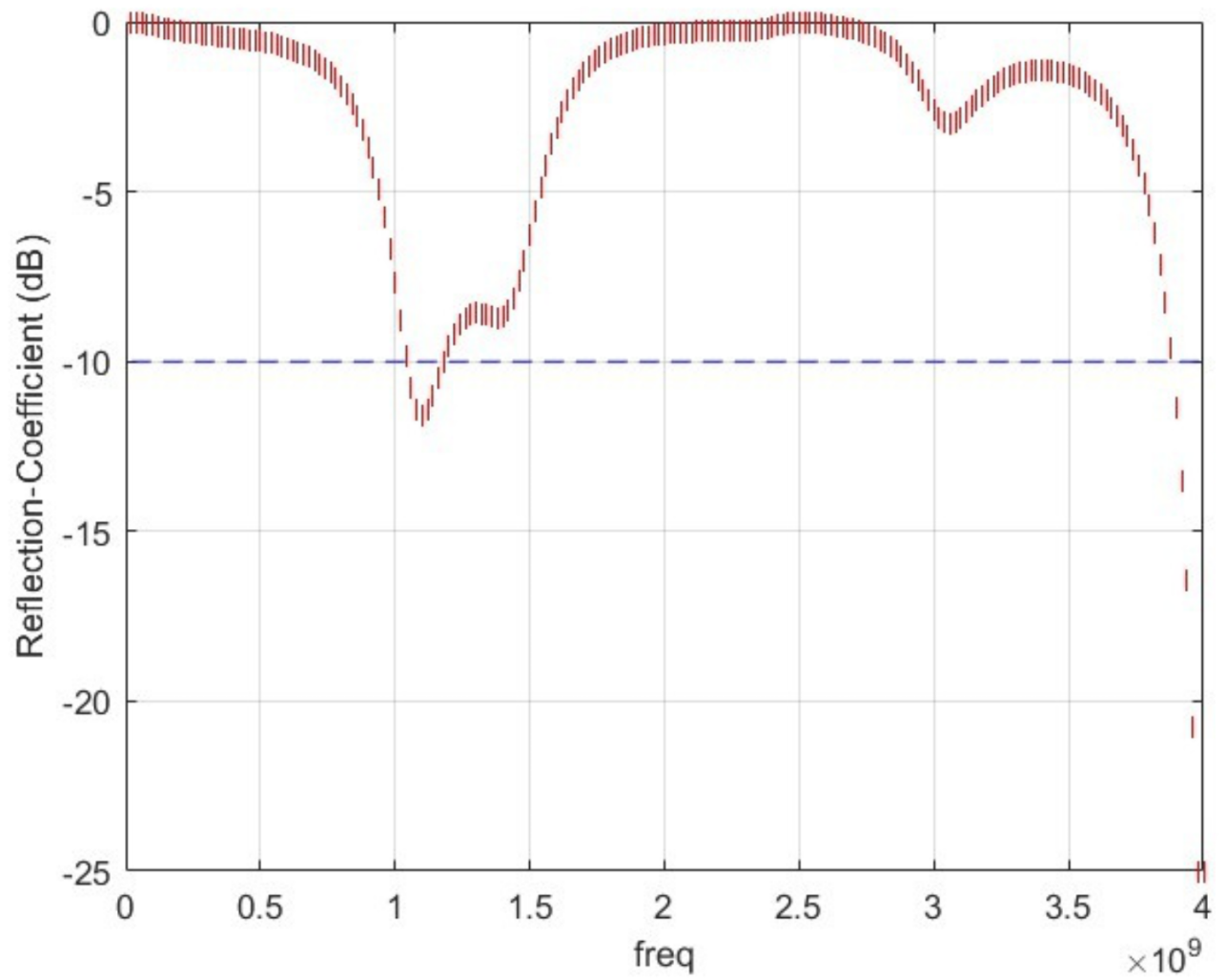


1.



ΓΡΑΦΗΜΑ ΜΕΤΡΟΥ ΣΥΝΤΕΛΕΣΤΗ ΑΝΑΚΛΑΣΗΣ $|\Gamma|$ ΣΕ
ΚΑΘΑΡΟ ΑΡΙΘΜΟ.

2.



ΓΡΑΦΗΜΑ ΣΥΝΤΕΛΕΣΤΗ ΑΝΑΚΛΑΣΗΣ ΣΕ dB.